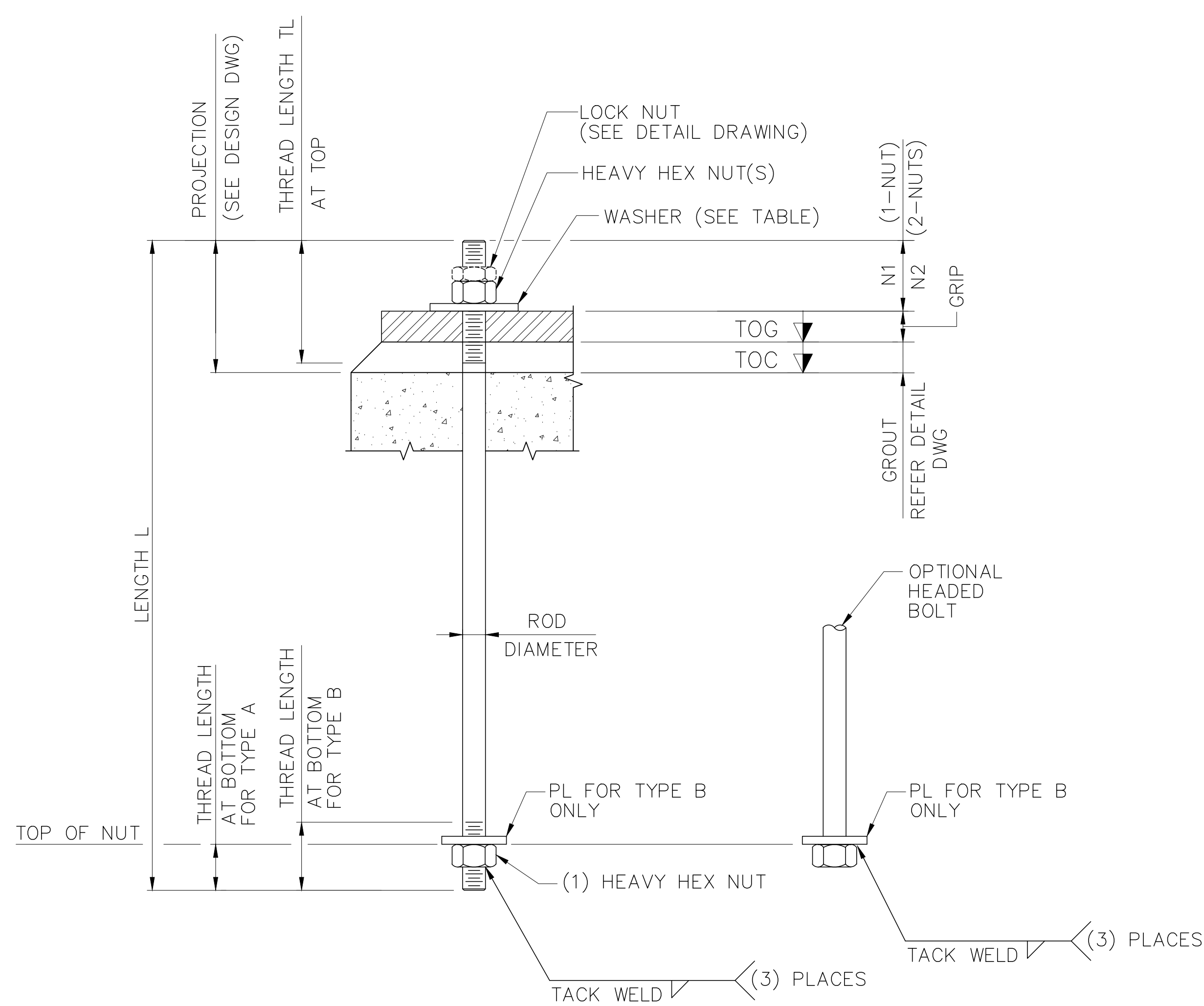
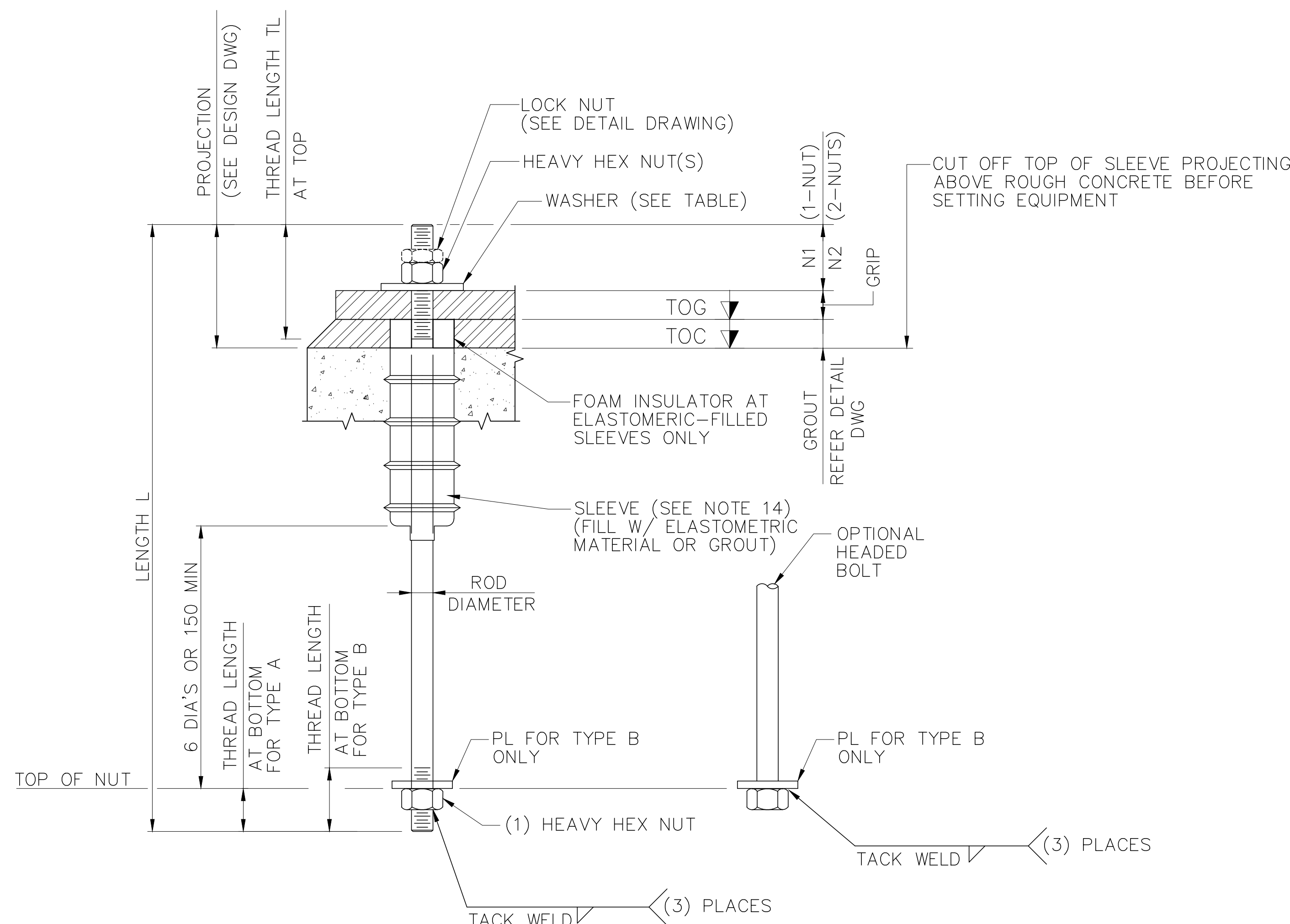




TYPES A, B AND N* ANCHOR ROD DETAILS

FOR DIMENSIONS, SEE ANCHOR ROD DATA TABLE
PROJECTION AND GROUT THICKNESS ARE AS SPECIFIED ON THE DESIGN DRAWINGS
* SEE SPECIAL NOTE 3



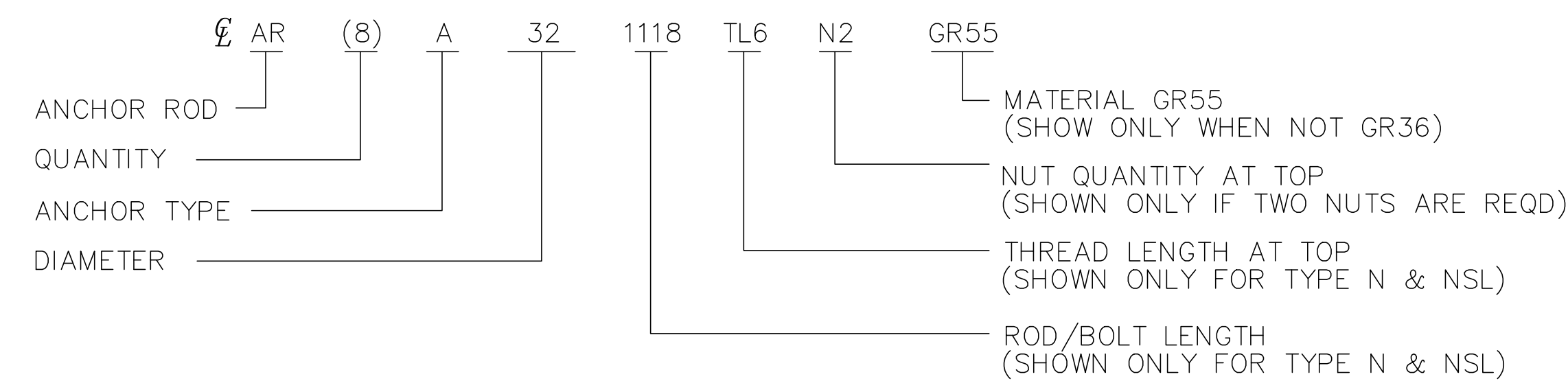
TYPES ASL, BSL AND NSL*
ANCHOR ROD DETAILS

FOR DIMENSIONS, SEE ANCHOR ROD DATA TABLE
PROJECTION AND GROUT THICKNESS ARE AS SPECIFIED ON THE DESIGN DRAWINGS
* SEE SPECIAL NOTE 3

DIAMETER	TYPE A ASL	TYPE B BSL			ALLOWANCE FOR NUTS		THREAD LENGTH			SLEEVE	WASHER TYPE	MINIMUM EDGE DISTANCE TO CONCRETE	MINIMUM SPACING BETWEEN BOLTS	NOMINAL STEEL STRENGTH OF ANCHOR BOLT IN IN TENSION, N _{sa} (kN) REFER NOTE 24	NOMINAL STEEL STRENGTH OF ANCHOR BOLT IN IN SHEAR, V _{sa} (kN) REFER NOTE 24
	LENGTH L	LENGTH L	PLATE SIZE		N1	N2	AT BOTTOM		AT TOP	SHELL SIZE					
			PLATE DIA	PLATE THK			FOR TYPE A	FOR TYPE B							
12	305	710	NR	NR	25	40	30	—	60	50 X 130	ASTM F436/F436M	150	150	24	12
16	380	735	NR	NR	30	50	30	—	70	50 X 180	ASTM F436/F436M	150	150	44	23
20	405	765	35	6	40	60	30	60	75	50 X 180	ASTM F436/F436M	150	125	72	37
22	435	790	45	6	40	65	40	65	85	50 X 180	ASTM F436/F436M	150	150	87	45
24	560	890	45	6	45	70	45	75	90	75 X 255	ASTM F436/F436M	150	140	102	53
30	610	1070	55	10	60	90	50	90	110	75 X 255	ASTM F436/F436M	150	150	165	85
36	815	1345	70	10	65	105	60	105	120	100 X 380	ASTM F436/F436M	175	165	239	124
42	865	1525	85	10	75	120	65	120	140	100 X 380	ASTM F436/F436M	200	200	338	175
48	1015	1575	90	12	85	135	70	135	155	100 X 460	ASTM F436/F436M	225	215	436	227
56	1065	1650	105	12	95	155	75	145	170	100 X 460	ASTM F436/F436M	275	240	614	319
64	1270	1700	115	12	105	165	90	165	185	155 X 610	ASTM F436/F436M	300	250	787	409
72	1320	1730	130	16	115	185	95	180	205	155 X 610	ASTM F436/F436M	325	275	1020	530
78	1370	1830	140	20	120	200	105	200	215	155 X 610	ASTM F436/F436M	375	275	1215	631

SPECIAL NOTES:

1. TYPE A & ASL – STANDARD LENGTH ANCHOR ROD
2. TYPE B & BSL – EXTRA LONG LENGTH ANCHOR ROD
3. TYPE N & NSL ANCHOR ROD HAS NON-STANDARD LENGTH OR THREAD LENGTH. SEE THE DESIGN DRAWING CALLOUT FOR DATA. SEE DATA TABLE FOR ALL OTHER INFORMATION.
4. NR = PLATE NOT REQUIRED FOR DUCTILITY
5. ALL DIMENSIONS ABOVE ARE IN mm.



ANCHOR ROD CALLOUT

(CAST-IN-PLACE)

MARABAN		Ma'aden Phosphate 3 – Phase 1	
NAME	TITLE	SIGN	DATE
REVIEW GRADE			
Result	Code	Description	
☐	A	Accepted	
☐	C	Proceed with exceptions to next Submission and Revision	
☐	R	Do Not Proceed - Revise as noted and Resubmit	
Acceptance in any of these categories in no way relieves the Contractor/Supplier of its responsibility for the due and proper performance of the Works in accordance with the Contract/Purchase Order.			
PROJECT MANAGER SIGN-OFF:			
NAME			DATE

ISSUED FOR CONSTRUCTION

GENERAL NOTES:

- 1) FOR CONCRETE GENERAL NOTES AND ABBREVIATIONS SEE DRAWING MD-520-1000-EG-CV-STD-0001.
- 2) ALL MATERIALS, FABRICATION AND INSTALLATION SHALL CONFORM TO PIP ST05121 FABRICATION AND INSTALLATION OF ANCHOR BOLTS.
- 3) THE TERM ANCHOR ROD EQUALS ANCHOR BOLT IN PIP SPEC AND REFERS TO BOTH THREADED RODS AND OPTIONAL HEADED BOLTS.
- 4) ANCHOR RODS SHALL NOT BE REPAIRED, REPLACED OR FIELD MODIFIED WITHOUT THE APPROVAL OF STRUCTURAL ENGINEER OF RECORD. INSTALLATION TOLERANCES SHALL CONFORM TO PIP ST05121 AND SPECIFICATION NUMBER MD-520-1000-EG-CV-SPC-0002. ANCHOR BOLTS SHALL BE IN ACCORDANCE WITH MD-101-SMEM-EG-CV-SPC-0002 AND MD-101-SMEM-EG-ST-SPC-0002.
- 5) MATERIAL FOR ANCHOR RODS SHALL CONFORM TO ASTM F1554, GRADE 36 WITH UNC-2A THREADS, UNLESS NOTED.
- 6) MATERIAL & DETAIL FOR OTHER HIGH STRENGTH ANCHOR RODS, IF REQUIRED, SHALL BE SHOWN ON THE DESIGN DRAWING.
- 7) NUTS SHALL CONFORM TO ASTM A563/A563M, GRADE A, HEAVY HEX WITH UNC-2B THREADS.
- 8) ONE WASHER IS REQUIRED FOR EACH ANCHOR ROD. WASHERS SHALL BE HARDENED WASHERS CONFORMING TO ASTM F436/F436M.
- 9) PROVIDE (1) HEAVY HEX NUT WITH WASHER AT TOP OF ANCHOR ROD. UNLESS NOTED.
- 10) ANCHOR RODS, NUTS, AND WASHERS SHALL BE HOT-DIP GALVANIZED ACCORDING TO ASTM A153M, UNLESS NOTED.
- 11) UNLESS OTHERWISE NOTED IN THE DESIGN DRAWINGS, ALL ANCHOR RODS SHALL HAVE UNIFIED COARSE THREAD SERIES (UNC) THREADS. THE THREADS SHALL CONFORM TO ANSI 31.1 HAVING A CLASS 2A TOLERANCE FOR BOLTS, AND A CLASS 2B TOLERANCE FOR NUTS.
- 12) JAM BOTTOM NUT SECURELY THEN TACK WELD IN SHOP WITH 3mm FILLET (3 PLACES) TO PREVENT RELEASE OR BACKING OFF.
- 13) WELDING SHALL CONFORM TO AWS D1.1 USING CLASS E70-XX ELECTRODES.
- 14) SLEEVES SHALL BE WILSON PLASTIC ANCHOR BOLT SLEEVES AS MANUFACTURED BY WILSON ANCHOR SLEEVE INC. (OR APPROVED EQUIVALENT).
- 15) VENDOR SHALL ATTACH A WEATHERPROOF TAG FOR IDENTIFICATION OF ALL ANCHOR RODS ACCORDING TO THEIR MARK DESIGNATION.
- 16) ANCHOR ROD SLEEVES FOR STATIONARY EQUIPMENT AND STRUCTURAL BASE PLATES SHALL BE FILLED WITH CEMENTATION GROUT AFTER THE COLUMN OR EQUIPMENT IS IN PLACE AND THE ANCHOR RODS ARE ALIGNED. WATER OR OTHER LOOSE PARTICLES SHALL NOT BE ALLOWED TO COLLECT IN THE SLEEVE BEFORE THE SLEEVE IS GROUTED. (UNLESS NOTED OTHERWISE ON THE DESIGN DRAWINGS).
- 17) WHERE PRETENSIONED ANCHOR RODS WITH SLEEVES ARE SPECIFIED ON THE DESIGN DRAWINGS, THE SLEEVES SHALL BE FILLED WITH A NON-BINDING ELASTOMERIC MATERIAL.
- 18) ANCHOR ROD THREADS SHALL BE COVERED WITH DUCT TAPE OR OTHER SUITABLE MEANS TO KEEP THEM CLEAN AND TO PREVENT ANY DAMAGE THAT MIGHT OCCUR DURING THE PREPARATION OF THE FOUNDATION FOR GROUTING AND THE ACTUAL GROUTING OF THE FOUNDATION.
- 19) UNLESS TORQUE VALUE IS SPECIFIED ON THE DESIGN DRAWINGS OR AT THE LOCATIONS OF SLIDE PLATES, ANCHOR RODS SHALL BE TIGHTENED TO A SNUG CONDITION. AT SLIDE PLATE LOCATIONS, NUTS SHALL BE BACKED OFF A HALF TURN AND A DOUBLE NUT SHALL BE ADDED.
- 20) WHERE (5) OR MORE ANCHOR RODS ARE REQUIRED FOR VERTICAL VESSEL OR STACKS, ANCHOR RODS SHALL BE HELD IN PLACE WITH A TEMPLATE AND SET ACCURATELY TO THE DIMENSIONS CALLED FOR ON THE FOUNDATION DRAWINGS.
- 21) GROUT SHALL BE IN ACCORDANCE WITH MD-520-1000-EG-ST-SPC-0003.
- 22) THE ULTIMATE BOLT CAPACITIES PROVIDED IN THE TABLE-1 ARE BASED ON THE FOLLOWING CONSIDERATIONS:
 - (i) SPECIFIED YIELD STRENGTH OF ANCHOR BOLT = 250 N/mm^2
 - (ii) SPECIFIED TENSILE STRENGTH OF ANCHOR BOLT = 400 N/mm^2
- 23) TENSION AND SHEAR CAPACITIES PROVIDED IN THE TABLE ARE BASED ON ULTIMATE STRENGTH OF MATERIALS AND MUST NOT BE USED IN CONJUNCTION WITH NORMAL/UNFACTORED LOADS.
- 24) CAPACITIES OF THE ANCHOR BOLTS ARE GIVEN AS PER THEIR STEEL STRENGTHS IN WORKING LOADS. THE BOLT CAPACITIES AS PER CODE OF PRACTICE OUT FAILURES WILL BE COMPUTED BY THE ENGINEER DURING DETAILED DESIGN AND BOLT NUMBERS IN FOUNDATION WILL BE PROVIDED ACCORDINGLY.

REFERENCE SPECIFICATIONS:

1. MD-520-1000-EG-CV-SP-0002 CONCRETE WORK SPECIFICATION.
2. MD-520-1000-EG-CV-CRT-0001 CIVIL AND STRUCTURAL DESIGN CRITERIA - COMMON.
3. MD-520-1000-EG-ST-SP-0001 STRUCTURAL STEEL CONSTRUCTION.
4. PIP STS03001 REINFORCED CONCRETE SPECIFICATION.
5. PIP STS05121 ANCHOR BOLT DESIGN GUIDE.
6. PIP STS05121 FABRICATION AND INSTALLATION OF ANCHOR BOLTS.
7. PIP STS03600 NONSHRINK CEMENTITIOUS GROUT SPECIFICATION.
8. PIP STS03601 EPOXY GROUT SPECIFICATION.

[illegible]